Dormfix - Test Cases and Results

Hostel Complaint & Maintenance Management System - Mid-Semester Evaluation

Group: [Group Number]

1. Test approach and environment

Backend behaviour is tested automatically with **Jest 29** and **Supertest**: unit tests for pure logic (overdue rule, JWT, role middleware, response helpers) and integration tests that call the real Express app over HTTP against a real MongoDB database that is wiped before every test. Every automated test carries its Test ID in its name (e.g. *TC-24: ...*) and records the observed HTTP status/message, so the **Actual Result** columns below are generated from the run, not typed by hand. Browser behaviour (forms, navigation, role-based screens) is covered by the manual UI cases in section 4, which must be run and screenshotted by a group member.

Run date (UTC)	2026-10-17T04:33:08.031Z
Node.js	v22.22.2
Platform	Linux 6.18.44-fc-v139 (x64)
Database used	MongoDB-compatible server (FerretDB 1.24.0, SQLite backend) - run inside the AI sandbox
Command	npm run test:evidence
Test target	04_Source_Code/source_files/server (submission snapshot)

2. Summary

Module	Cases	Pass	Fail
Authentication	18	18	0
Authorization	7	7	0
Complaint submission	7	7	0
Complaint retrieval	9	9	0
Status workflow	9	9	0
Overdue logic	5	5	0
Feedback	8	8	0
Dashboard	3	3	0
Utilities (unit)	4	4	0
Automated total	70	70	0
Manual UI (pending until executed)	21	0	0 fail / 21 pending

Code coverage of the backend (Jest/Istanbul): statements 93.86%, branches 80.48%, functions 92.06%, lines 94.14%.

3. Automated test cases (Jest + Supertest)

ID	Module	Test description	Input / scenario	Expected result	Actual result (recorded)	Status
TC-01	Authentication	register a new resident with valid details	POST /api/auth/register with valid name, unique reg. no., room, email, password	201 Created; JWT returned; role = resident; password not in response	HTTP 201 - Registration successful.; token returned: true; role=resident	PASS
TC-02	Authentication	register with a duplicate email	Register with an email that already exists	409; "Email is already registered."	HTTP 409 - Email is already registered.	PASS
TC-03	Authentication	register with a duplicate registration number	Register with a registration number already in use	409; "Registration number is already in use."	HTTP 409 - Registration number is already in use.	PASS
TC-04	Authentication	register with a password shorter than 6 characters	Register with password "123"	400; message about minimum 6 characters	HTTP 400 - Password must be at least 6 characters.	PASS
TC-05	Authentication	login with correct credentials	POST /api/auth/login with correct credentials	200 OK; JWT returned	HTTP 200 - Login successful.; token returned: true	PASS
TC-06	Authentication	login with an incorrect password	Login with correct email, wrong password	401; generic "Invalid email or password."	HTTP 401 - Invalid email or password.	PASS
TC-07	Authentication	access a protected route with no token	GET /api/auth/me with no Authorization header	401; "Not authorized. No token provided."	HTTP 401 - Not authorized. No token provided.	PASS
TC-08	Authentication	access a protected route with a malformed/invalid token	GET /api/auth/me with token "not.a.jwt"	401; "Not authorized. Invalid or expired token."	HTTP 401 - Not authorized. Invalid or expired token.	PASS
TC-33	Authentication	client-supplied role is ignored on registration (no privilege escalation)	Register with extra field role = "warden"	201; role assigned is still "resident"	HTTP 201 - Registration successful.; role assigned = resident	PASS
TC-34	Authentication	register with a missing required field	Register with roomNumber missing	400; "All fields are required."	HTTP 400 - All fields are required.	PASS
TC-35	Authentication	register with an invalid email format	Register with email "not-an-email"	400; "Invalid email format."	HTTP 400 - Invalid email format.	PASS
TC-36	Authentication	login with an unregistered email gives the same generic error	Login with an email that is not registered	401; same generic message as TC-06 (no user enumeration)	HTTP 401 - Invalid email or password.	PASS
TC-37	Authentication	GET /api/auth/me with a valid token returns the user without password	GET /api/auth/me with a valid token	200; user returned; no password field	HTTP 200 - Current user fetched.; email=resident3@dormfix.test; password field present: false	PASS
TC-38	Authentication	password is stored hashed, never as plain text	Register, then read the stored user from the database	Stored password is a bcrypt hash, not the plain text	stored value starts with "$2a$" (bcrypt); equals plaintext: false	PASS
TC-39	Authentication	seeded warden can log in and receives role 'warden'	Login as a warden account	200; role = warden	HTTP 200 - Login successful.; role=warden	PASS
TC-40	Authentication	unknown API route returns 404 in the standard error shape	GET /api/does-not-exist	404 in standard error shape (success = false)	HTTP 404 - Route not found: /api/does-not-exist	PASS
TC-61	Authentication	token for a deleted user is rejected	Use a valid token after the user was deleted	401 Unauthorized	HTTP 401 - Not authorized. User no longer exists.	PASS
TC-62	Authentication	an expired token is rejected	Use a token that has already expired	401; "Not authorized. Invalid or expired token."	HTTP 401 - Not authorized. Invalid or expired token.	PASS
TC-10	Authorization	resident cannot access the warden-only complaint list	Resident calls GET /api/complaints (warden-only)	403 Forbidden	HTTP 403 - Forbidden. Role 'resident' cannot access this resource.	PASS
TC-11	Authorization	resident cannot view another resident's complaint	Resident calls GET /api/complaints/:id of another resident	403 Forbidden	HTTP 403 - Forbidden. This complaint does not belong to you.	PASS
TC-12	Authorization	resident cannot update complaint status	Resident calls PATCH /api/complaints/:id/status	403 Forbidden; status unchanged in DB	HTTP 403 - Forbidden. Role 'resident' cannot access this resource.	PASS
TC-13	Authorization	resident cannot reorder complaints	Resident calls PATCH /api/complaints/reorder	403 Forbidden	HTTP 403 - Forbidden. Role 'resident' cannot access this resource.	PASS
TC-14	Authorization	warden can list all complaints	Warden calls GET /api/complaints with 2 complaints in DB	200 OK; both complaints returned	HTTP 200 - Complaints fetched.; complaints returned: 2	PASS

ID	Module	Test description	Input / scenario	Expected result	Actual result (recorded)	Status
TC-43	Authorization	warden cannot submit a complaint (resident-only)	Warden calls POST /api/complaints	403 Forbidden (resident-only)	HTTP 403 - Forbidden. Role 'warden' cannot access this resource.	PASS
TC-63	Authorization	unauthenticated request to create a complaint is rejected	POST /api/complaints with no token	401 Unauthorized	HTTP 401 - Not authorized. No token provided.	PASS
TC-15	Complaint submission	submit a valid complaint without an image	Resident submits description (>= 5 chars), no image	201; status = SUBMITTED; complaintId = HF-xxxx; imagePath = null	HTTP 201 - Complaint submitted successfully.; status=SUBMITTED; complaintId=HF-0001; imagePath=null	PASS
TC-16	Complaint submission	submit a complaint with a valid image	Resident submits description + valid PNG image	201; imagePath = /uploads/<file>.png	HTTP 201 - Complaint submitted successfully.; imagePath=/uploads/complaint-1792211575805-865560279.png	PASS
TC-17	Complaint submission	description shorter than 5 characters is rejected	Description = "Hi" (2 chars)	400; nothing saved	HTTP 400 - Description is required and must be at least 5 characters.	PASS
TC-18	Complaint submission	image larger than 5 MB is rejected	Image larger than 5 MB	400; "Image file is too large."	HTTP 400 - Image file is too large.	PASS
TC-19	Complaint submission	unsupported file type (PDF) is rejected	PDF file sent as the image	400; "Invalid file type..."	HTTP 400 - Invalid file type. Only JPG, PNG and WEBP images are allowed.	PASS
TC-41	Complaint submission	complaint IDs are sequential (HF-0001, HF-0002)	Two residents submit one complaint each	IDs are HF-0001 then HF-0002	HF-0001, HF-0002	PASS
TC-42	Complaint submission	resident identity comes from the profile, not the request body	Body contains residentName, roomNumber, status = RESOLVED	Values ignored; profile data used; status = SUBMITTED	HTTP 201 - Complaint submitted successfully.; residentName=Resident 40; roomNumber=A-140; status=SUBMITTED	PASS
TC-20	Complaint retrieval	resident sees only their own complaint history	Resident calls GET /api/complaints/my (2 own, 1 foreign in DB)	200; only the 2 own complaints	HTTP 200 - Complaints fetched.; returned 2, all own: true	PASS
TC-44	Complaint retrieval	warden filters by status	Warden: GET /api/complaints?status=UNDER_PROGRESS	Only UNDER_PROGRESS complaints returned	HTTP 200 - Complaints fetched.; returned 1	PASS
TC-45	Complaint retrieval	warden filters overdue complaints only	Warden: GET /api/complaints?overdue=true (1 overdue, 1 fresh, 1 old-but-resolved)	Only the overdue unresolved complaint	HTTP 200 - Complaints fetched.; overdue returned 1	PASS
TC-46	Complaint retrieval	warden filters by room number	Warden: GET /api/complaints?roomNumber=<room>	Only that room's complaints	HTTP 200 - Complaints fetched.; returned 1	PASS
TC-47	Complaint retrieval	invalid status filter is rejected	Warden: GET /api/complaints?status=BOGUS	400 Bad Request	HTTP 400 - Invalid status filter.	PASS
TC-48	Complaint retrieval	sorting by oldest / newest orders by creation time	Warden: sortBy=oldest and sortBy=newest	Correct chronological ordering in both cases	oldest-first order correct: true; newest-first order correct: true	PASS
TC-50	Complaint retrieval	malformed complaint ID returns 400	GET /api/complaints/not-an-id	400; "Invalid ID format."	HTTP 400 - Invalid ID format.	PASS
TC-51	Complaint retrieval	well-formed but non-existent complaint ID returns 404	GET /api/complaints/000000000000000000000000	404 Not Found	HTTP 404 - Complaint not found.	PASS
TC-64	Complaint retrieval	overdue flag is recalculated and persisted when complaints are read	Read a 70-hour-old unresolved complaint stored with isOverdue = false	Response and database both show isOverdue = true	API isOverdue=true; persisted=true	PASS
TC-21	Status workflow	SUBMITTED -> UNDER_PROGRESS	Warden: SUBMITTED -> UNDER_PROGRESS	200; status updated	HTTP 200 - Status updated.; status=UNDER_PROGRESS	PASS
TC-22	Status workflow	UNDER_PROGRESS -> RESOLVED sets resolvedAt and clears overdue	Warden: UNDER_PROGRESS -> RESOLVED (complaint was overdue)	200; resolvedAt set; isOverdue cleared	HTTP 200 - Status updated.; status=RESOLVED; resolvedAt set=true; isOverdue=false	PASS
TC-23	Status workflow	RESOLVED -> SUBMITTED is rejected (terminal state)	Warden: RESOLVED -> SUBMITTED	400; "Invalid status transition..."	HTTP 400 - Invalid status transition from RESOLVED to SUBMITTED.	PASS

ID	Module	Test description	Input / scenario	Expected result	Actual result (recorded)	Status
TC-52	Status workflow	SUBMITTED -> RESOLVED directly is allowed (SRS BR-05)	Warden: SUBMITTED -> RESOLVED directly	200 OK (allowed by SRS BR-05)	HTTP 200 - Status updated.; status=RESOLVED	PASS
TC-49	Status workflow	unknown status value in the request is rejected	PATCH status = "DONE"	400 Bad Request	HTTP 400 - A valid status is required.	PASS
TC-65	Status workflow	UNDER_PROGRESS -> SUBMITTED (backwards) is rejected	Warden: UNDER_PROGRESS -> SUBMITTED	400; invalid transition	HTTP 400 - Invalid status transition from UNDER_PROGRESS to SUBMITTED.	PASS
TC-24	Status workflow	warden moves a complaint up (priorityOrder swapped with neighbour)	Warden moves middle complaint "up"	200; priorityOrder swapped with the neighbour above; third unchanged	HTTP 200 - Complaint reordered.; a<->b swapped: true	PASS
TC-53	Status workflow	moving the top complaint up is a no-op	Warden moves the top complaint "up"	200; no change (already at top)	HTTP 200 - Complaint reordered.; priorityOrder unchanged: true	PASS
TC-54	Status workflow	reorder with an invalid direction is rejected	Reorder with direction = "sideways"	400 Bad Request	HTTP 400 - Direction must be 'up' or 'down'.	PASS
TC-25	Overdue logic	complaint created 5 hours ago is not overdue	Complaint created 5 h ago, SUBMITTED	isOverdue = false	isOverdue = false	PASS
TC-26	Overdue logic	unresolved complaint older than 48 hours is overdue	Complaint created 60 h ago, UNDER_PROGRESS	isOverdue = true	isOverdue = true	PASS
TC-27	Overdue logic	resolved complaint older than 48 hours is never overdue	Complaint created 80 h ago, RESOLVED	isOverdue = false	isOverdue = false	PASS
TC-55	Overdue logic	boundary - 47h59m not overdue, 48h01m overdue	Boundary: 47.98 h and 48.02 h old, SUBMITTED	false at 47.98 h; true at 48.02 h	47.98h -> false; 48.02h -> true	PASS
TC-56	Overdue logic	threshold constant equals 48 hours in milliseconds	Read OVERDUE_THRESHOLD_MS	Equals 48 hours in milliseconds	OVERDUE_THRESHOLD_MS = 172800000	PASS
TC-28	Feedback	submit feedback on a resolved complaint	Resident rates a resolved complaint 4 with a comment	201; feedback stored; complaint.feedbackSubmitted = true	HTTP 201 - Feedback submitted.; rating=4; complaint.feedbackSubmitted=true	PASS
TC-29	Feedback	feedback on an unresolved complaint is rejected	Feedback on a SUBMITTED complaint	400; "Feedback can only be submitted after the complaint is resolved."	HTTP 400 - Feedback can only be submitted after the complaint is resolved.	PASS
TC-30	Feedback	second feedback on the same complaint is rejected	Submit feedback twice on the same complaint	Second attempt 409; only 1 feedback document stored	HTTP 409 - Feedback has already been submitted for this complaint.; feedback docs in DB: 1	PASS
TC-31	Feedback	rating outside 1-5 is rejected	rating = 7 and rating = 0	400; "Rating must be between 1 and 5."	rating 7 -> HTTP 400 - Rating must be between 1 and 5.; rating 0 -> HTTP 400 - Rating must be between 1 and 5.	PASS
TC-32	Feedback	resident cannot give feedback on another resident's complaint	Resident rates another resident's complaint	403 Forbidden	HTTP 403 - Forbidden. You can only give feedback on your own complaints.	PASS
TC-66	Feedback	comment longer than 500 characters is rejected	Comment of 501 characters	400 Bad Request	HTTP 400 - Validation failed.	PASS
TC-67	Feedback	warden can read feedback; resident can read own; other resident cannot	GET feedback as warden, as owner, as another resident	200, 200, 403	warden 200, owner 200, other resident 403	PASS
TC-68	Feedback	warden cannot submit feedback (resident-only)	Warden POSTs feedback	403 Forbidden (resident-only)	HTTP 403 - Forbidden. Role 'warden' cannot access this resource.	PASS
TC-69	Dashboard	counts, overdue total and average resolution time are correct	Warden: 2 submitted (1 overdue), 1 under progress, 2 resolved (10 h and 30 h)	total 5, submitted 2, underProgress 1, resolved 2, overdue 1, avg resolution about 20 h, 14 day entries	HTTP 200 - Dashboard stats fetched.; total=5, submitted=2, underProgress=1, resolved=2, overdue=1, avgResolutionHours=20, resolvedPerDay entries=14	PASS
TC-70	Dashboard	dashboard with no complaints returns zeros and null average	Warden dashboard with an empty database	All counts 0; avgResolutionHours = null	HTTP 200 - Dashboard stats fetched.; total=0; avgResolutionHours=null	PASS

ID	Module	Test description	Input / scenario	Expected result	Actual result (recorded)	Status
TC-71	Dashboard	resident cannot access the warden dashboard	Resident calls GET /api/dashboard/warden	403 Forbidden	HTTP 403 - Forbidden. Role 'resident' cannot access this resource.	PASS
TC-57	Utilities (unit)	JWT contains user id and role and an expiry	generateToken({_id, role})	Token payload has id, role and an expiry	payload keys: id, role, iat, exp; role=resident	PASS
TC-58	Utilities (unit)	authorize() blocks a role that is not allowed (403)	authorize("warden") with a resident user	403 and next() not called	status 403; next called: false	PASS
TC-59	Utilities (unit)	authorize() allows a permitted role and calls next()	authorize("warden") with a warden user	next() called	next called: true	PASS
TC-60	Utilities (unit)	success()/error() helpers return the standard response shape	success() and error() helpers	Standard {success, message, data/errors} shape	success -> {"success":true,"message":"ok","data":{"x":1}}; error -> {"success":false,"message":"bad","errors":["e1"]}	PASS

4. Manual UI test cases (to be executed in the browser)

Precondition: server and client running, database seeded with *npm run seed*. Save one screenshot per case as **Test_Evidence/<ID>.png** and record the result in **manual_results.csv**, then re-run *generate_test_docs.py*.

ID	Area	Steps	Expected result	Actual result	Evidence file	Status
TC-09	Authentication (UI)	Log in, then click "Logout" in the navbar.	Redirected to /login; browser localStorage no longer has dormfix_token / dormfix_user.	Not yet executed - run manually and record result	TC-09.png (to capture)	PENDING
UI-01	Authentication (UI)	Open /register, fill all fields with valid new data, click the register button.	Account created; redirected to /resident/dashboard; navbar shows your name and "(Resident)".	Not yet executed - run manually and record result	UI-01.png (to capture)	PENDING
UI-02	Authentication (UI)	On /register enter password "123" and submit.	Red alert: "Password must be at least 6 characters."	Not yet executed - run manually and record result	UI-02.png (to capture)	PENDING
UI-03	Authentication (UI)	Register again with an email that already exists.	Red alert showing the server message ("Email is already registered.").	Not yet executed - run manually and record result	UI-03.png (to capture)	PENDING
UI-04	Authentication (UI)	Log in as resident aarav@dormfix.edu (seed data).	Redirected to /resident/dashboard.	Not yet executed - run manually and record result	UI-04.png (to capture)	PENDING
UI-05	Authentication (UI)	Log in with a correct email and a wrong password.	Red alert "Invalid email or password."; stays on /login.	Not yet executed - run manually and record result	UI-05.png (to capture)	PENDING
UI-06	Authentication (UI)	Log in as warden@dormfix.edu (seed data).	Redirected to /warden/dashboard; navbar shows "(Warden)".	Not yet executed - run manually and record result	UI-06.png (to capture)	PENDING
UI-07	Access control (UI)	(a) Logged out, open /warden/dashboard. (b) Logged in as resident, open /warden/dashboard.	(a) Redirected to /login. (b) Redirected away to the resident area; warden page not shown.	Not yet executed - run manually and record result	UI-07.png (to capture)	PENDING
UI-08	Complaints (UI)	As a resident open "New Complaint", enter a description of 10+ characters, no image, submit.	Success; complaint appears on the resident dashboard with status SUBMITTED and an HF-xxxx ID.	Not yet executed - run manually and record result	UI-08.png (to capture)	PENDING
UI-09	Complaints (UI)	Submit a complaint with a JPG/PNG under 5 MB. Open it as the resident, then as the warden.	Image preview shown before submit; image visible on both detail pages.	Not yet executed - run manually and record result	UI-09.png (to capture)	PENDING
UI-10	Complaints (UI)	Enter description "Hi" and submit.	Red alert "Description must be at least 5 characters."; no request sent.	Not yet executed - run manually and record result	UI-10.png (to capture)	PENDING
UI-11	Complaints (UI)	Choose a PDF file, then a file over 5 MB.	"Only JPG, PNG, and WEBP images are allowed." then "Image must be smaller than 5MB."	Not yet executed - run manually and record result	UI-11.png (to capture)	PENDING
UI-12	Complaints (UI)	Log in as aarav@dormfix.edu and open the dashboard.	Only Aarav's complaints listed, each with a status badge; none from other residents.	Not yet executed - run manually and record result	UI-12.png (to capture)	PENDING
UI-13	Warden (UI)	Log in as warden; open the dashboard.	Cards: Total, Submitted, Under Progress, Resolved, Overdue, Avg. Resolution (hrs), plus the 14-day chart; numbers match the table.	Not yet executed - run manually and record result	UI-13.png (to capture)	PENDING
UI-14	Warden (UI)	Use the Status, Overdue, Room Number and Sort By filters one at a time.	Table updates to match every filter; overdue rows show the red "Overdue" badge.	Not yet executed - run manually and record result	UI-14.png (to capture)	PENDING
UI-15	Warden (UI)	With Sort By = priority click the up arrow on a complaint, then the down arrow.	Row moves one place up, then back down. (Arrows are hidden for other sort modes.)	Not yet executed - run manually and record result	UI-15.png (to capture)	PENDING
UI-16	Warden (UI)	Open a SUBMITTED complaint; click "Mark as UNDER PROGRESS", then "Mark as RESOLVED".	Badge changes after each click; once RESOLVED no further status buttons appear.	Not yet executed - run manually and record result	UI-16.png (to capture)	PENDING
UI-17	Feedback (UI)	As the resident who owns a RESOLVED complaint, open it, give a rating and comment, submit.	Feedback saved and shown read-only; form is no longer offered.	Not yet executed - run manually and record result	UI-17.png (to capture)	PENDING
UI-18	Feedback (UI)	As a resident open a complaint that is not yet RESOLVED.	No feedback form is shown.	Not yet executed - run manually and record result	UI-18.png (to capture)	PENDING
UI-19	Access control (UI)	As resident A, paste the URL /resident/complaints/<ID of resident B's complaint>.	Access denied / error message; complaint details not displayed.	Not yet executed - run manually and record result	UI-19.png (to capture)	PENDING

ID	Area	Steps	Expected result	Actual result	Evidence file	Status
UI-20	Known issue (UI)	On /login click "Register as Warden".	Expected per SRS: wardens are seeded, not self-registered. Actual: link goes to /warden/register, which has no route. Record the result and fix (see report).	Not yet executed - run manually and record result	UI-20.png (to capture)	PENDING

5. Observations and defects found during testing

ID	Severity	Observation	Recommended fix
OBS-1	High	Live app folder (server/server.js): the feedback and dashboard routes are commented out, so feedback (TC-28 to TC-32, TC-66 to TC-68) and the warden dashboard statistics (TC-69 to TC-71) return 404 there. The submission snapshot (04_Source_Code) has them enabled and all of those tests pass. Running the same suite against the live folder gave 59 pass / 11 fail; un-commenting the four lines gave 70/70.	Un-comment the two require() lines and the two app.use() lines in server/server.js, or re-sync the live code from 04_Source_Code.
OBS-2	Medium	Login page offers a "Register as Warden" link to /warden/register, but that route does not exist in App.jsx (the catch-all redirects to /), and the SRS says wardens are created by the seed script only.	Remove the link from Login.jsx (see UI-20).
OBS-3	Low	Mongoose prints "Duplicate schema index" warnings for email and registrationNumber in models/User.js: the fields use unique: true and the file also calls schema.index() for them.	Delete the two userSchema.index(...) lines.
OBS-4	Info	The original tests/ folder contained only scaffolds with placeholder assertions (expect(true).toBe(true)) and could not run without a database. They were replaced by the suite documented here.	None.

6. How to reproduce

From *04_Source_Code/source_files/server*: **npm install**, then **npm test** (or **npm run test:evidence** to also save logs, JSON results and coverage under tests/evidence/). The first run downloads a small MongoDB binary for mongodb-memory-server (internet needed once). To use an existing MongoDB instead, set MONGO_URI_TEST to a throw-away database URI. No real credentials are used by the tests.